GT
10

Date: 22-Apr-2025

Sri Chaitanya Educational Institutions | Infinity Learn

FINAL PHASE REVISION

NEET GRAND TEST-10 [SET-1]

Sr Elite (P1)

Syllabus

Botany : Full Syllabus of NEET 2025

Zoology : Full Syllabus of NEET 2025

Physics : Full Syllabus of NEET 2025

Chemistry : Full Syllabus of NEET 2025

INSTRUCTIONS

- The Answer Sheet is inside this Test Booklet. When you are directed to open the Test Booklet, take out the Answer Sheet and fill in the particulars on side-1 and side-2 carefully with blue/black ball point pen only.
- The test is of 3 hours duration and Test Booklet contains 180 questions. Each question carries 4 marks. For each correct response, the candidate will get 4 marks. For each incorrect response, one mark will be deducted from the total scores. The maximum marks are 720.
- Use Blue/Black Ball Point Pen only for writing particulars on this page/marking responses.
- On completion of the test, the candidate must hand over the Answer Sheet to the invigilator before leaving the Room/Hall. The candidates are allowed to take away this Test Booklet with them.
- The candidates should ensure that the Answer Sheet is not folded. Do not make any stray marks on the Answer Sheet. Do not write your Roll No. anywhere else except in the specified space in the Test Booklet/Answer Sheet.
- Use of white fluid for correction, electronic/manual calculator are NOT permissible.
- Each candidate must show on demand his/her Admit Card to the Invigilator.
- No candidate, without special permission of the Superintendent or Invigilator, would leave his/her seat.
- The candidates should not leave the Examination Hall without handing over there Answer Sheet to the Invigilator on duty and sign the Attendance Sheet twice. Cases where a candidate has not signed the Attendance Sheet second time will be deemed not to have handed over the Answer Sheet and dealt with as an unfair means case.
- The candidates are governed by all Rules and Regulations of the examination with regard to their conduct in the Examination Hall. All cases of unfair means will be dealt with as per Rules and Regulations of this examination.
- No part of the Test Booklet and Answer Sheet shall be detached under any circumstances.

CHEMISTRY

1. Entropy is maximum for 10 grams of
   (1) Methane (2) Water
   (3) Carbon (4) Phosphorous

2. Match List-I with List-II :

	List-I		List-II
A.	$\Delta_{total}S > 0$	P.	W_{useful} done by system
B.	$\Delta_{total}S < 0$	Q.	Spontaneous
C.	ΔH	R.	Non-spontaneous
D.	$-\Delta G$	S.	$\Delta U+\Delta n_g RT$

Choose the correct answer from the options given below :

	(A)	(B)	(C)	(D)
(1)	R	P	S	Q
(2)	S	Q	P	R
(3)	R	S	P	Q
(4)	Q	R	S	P

3. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

   Assertion (A) : Heterolytic cleavage of bromomethane will give CH_3^+ and Br^- as shown below.

   $H_3C-Br \longrightarrow H_3\overset{+}{C} + Br^-$

   Reason (R) : In *heterolytic cleavage,* one of the electrons of the shared pair in a covalent bond goes with each of the bonded atoms.

   In light of the above statements, choose the correct answer from the options given below.

   (1) Both (A) and (R) are true and (R) is the correct explanation of (A).
   (2) Both (A) and (R) are true and (R) is not the correct explanation of (A).
   (3) (A) is true but (R) is false.
   (4) (A) is false but (R) is true.

4. pH of a solution is 2 and that of another is 4. When equal volumes of these two are mixed, p^H of the resulting solution is
   (1) 2.3 (2) 3
   (3) 3.2 (4) 3.7

5. Which of the following aqueous solutions has the highest boiling point?
   (1) 0.1 M KNO_3 (2) 0.1 M Na_3PO_4
   (3) 0.1 M $BaCl_2$ (4) 0.1 M K_2SO_4

6. Potential energy of an electron present in an orbit of He^+ is –6.8 eV. The angular momentum of same electron revolving in the same orbit of He^+ is
   (1) $\frac{h}{\pi}$ (2) $\frac{h}{2\pi}$
   (3) $\frac{2h}{\pi}$ (4) $\frac{2\pi}{h}$

7. In $\psi_{(3,2,1)}$, the sum of orbital angular momentum, spherical nodes and angular nodes is
   (1) $\frac{\sqrt{6}h+4\pi}{2\pi}$
   (2) $\frac{\sqrt{6}h}{2\pi}+3$
   (3) $\frac{\sqrt{6}h+2\pi}{2\pi}$
   (4) $\frac{\sqrt{6}h+8\pi}{2\pi}$

8. Maximum number of planar atoms in SF_6 molecule
   (1) 5 (2) 4
   (3) 6 (4) 7

9. On strong heating, 2.76 g of Ag_2CO_3 yields a residue weighing

   $Ag_2CO_3 \longrightarrow 2Ag+\frac{1}{2}O_2+CO_2$

   (1) 3.48 g (2) 1.44 g
   (3) 2.16 g (4) 4.16 g

10. Match List-I with List -II :

	List-I (Reactions)		List-II (Reagents)
A.	Ethane nitrile to ethanal	**p.**	PCC
B.	Allyl alcohol to propenal	**q.**	DIBAL-H
C.	But-2-ene to ethanal	**r.**	$O_3 Zn / H_2O$

Choose the correct answer from the options given below :

	(A)	(B)	(C)
(1)	q	p	r
(2)	r	q	p
(3)	r	p	q
(4)	q	r	p

11. Which of the following ions does ***not*** exist in an aqueous solution?

(1) Pb^{2+} (2) Sn^{4+}

(3) B^{3+} (4) Tl^{+}

12. In carbon family the element which can ***not*** show catenation

(1) Carbon (2) Germanium

(3) Silicon (4) Lead

13. Exothermic and endothermic hydrides of 16th group are respectively

(1) H_2O, H_2S (2) H_2Se, H_2O

(3) H_2Te, H_2S (4) H_2S, H_2Se

14. Arrange the following resonating structures of vinyl chloride in order of decreasing stability.

$$\underset{\text{I}}{H_2C = CH - Cl} \longleftrightarrow \underset{\text{II}}{H_2\bar{C} - CH = \overset{+}{Cl}} \longleftrightarrow \underset{\text{III}}{\overset{+}{H_2C} - CH = \bar{Cl}}$$

(1) I > II > III (2) II > I > III

(3) III > II > I (4) I > II = III

15. Given below are two statements :

Statement-I : Galvanic cell is example of a reversible cell.

Statement-II: In a galvanic cell the salt bridge helps to maintain electrical neutrality.

In light of the above statements, choose the correct answer from the options given below.

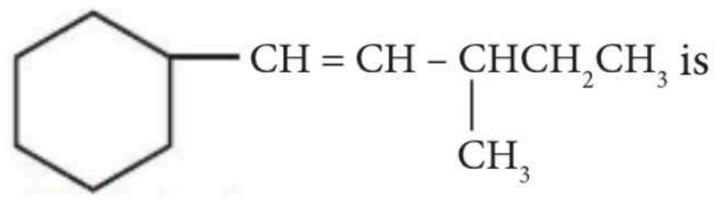
(1) Both statement I and statement II are correct.

(2) Statement I is incorrect but statement II is correct.

(3) Statement I is correct but statement II is incorrect.

(4) Both statement I and statement II are incorrect.

16. The IUPAC name of



(1) 3-methyl-5-cyclohexyl-pent-1-ene

(2) 1-cyclohexyl-3-methyl-1-pentene

(3) 1-cyclohexyl-3-ethyl-but-1-ene

(4) 1-cyclohexyl-3,4-dimethyl-but-1-ene

17. Which one of the following gives positive neutral $FeCl_3$ solution test?

(1) Phenol (2) Ethoxy Ethane

(3) Ethanal (4) Propanone

18. Given below are two statements :

Statement-I : Vapour pressure of 0.5 M sugar solution is more than 0.5 M KCl solution.

Statement-II: Relative lowering of vapour pressure is equal to mole fraction of non volatile solute particles in the solution.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is incorrect but statement II is correct.

(3) Statement I is correct but statement II is incorrect.

(4) Both statement I and statement II are incorrect.

19. How many faradays are required to reduce 1 mole of BrO_3^- to Br^-?

(1) 3 (2) 5

(3) 6 (4) 4

20. Calculate the half-life of the first-order reaction

$$C_2H_4O(g) \longrightarrow CH_4(g) + CO(g)$$

if the initial pressure of $C_2H_4O(g)$ is 80 mm and the total pressure at the end of 20 minutes is 120 mm.

(1) 40 min (2) 120 min

(3) 20 min (4) 80 min

21. In a chemical reaction A → B, it is found that the rate of the reaction doubles when the concentration of A is increased four times. The order of the reaction with respect to A is

(1) 0 (2) 1/2

(3) 1 (4) 2

22. The IUPAC name for the complex compound $Li[AlH_4]$ is

(1) lithium aluminium hydride

(2) hydrido aluminium lithium (III)

(3) lithium tetrahydridoaluminate (III)

(4) lithium tetrahydridoaluminate (I)

23. Which of the following statements is ***correct***?

(1) The $[Ni(CN)_4]^{2-}$ ion has tetrahedral geometry and is diamagnetic.

(2) The $[Ni(CN)_4]^{2-}$ ion has a square-planar geometry and is paramagnetic.

(3) The $[Ni(CN)_4]^{2-}$ ion has a square-planar geometry and is diamagnetic.

(4) The $[Cu(NH_3)_4]^{2+}$ ion has a tetrahedral geometry and is diamagnetic.

24. Which of the following types of square-planar complexes can show geometrical isomers.

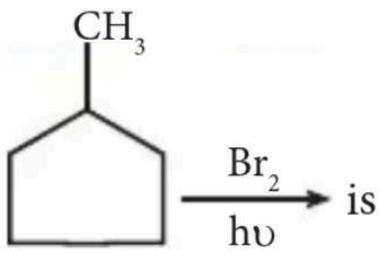
(1) Ma_4

(2) Ma_3b

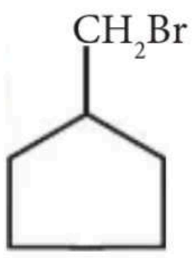
(3) Ma_2b_2

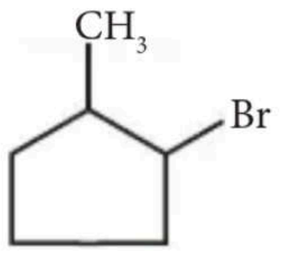
(4) Mab_3

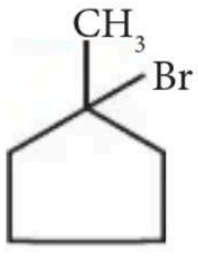
25. The major product obtained in the reaction

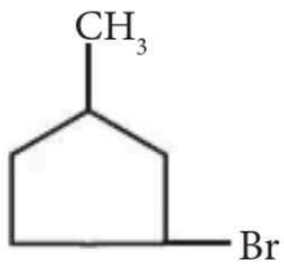


is

(1) 

(2) 

(3) 

(4) 

26. Arrange the following compounds in increasing order of boiling points

(1) n-butane, ethoxyethane, pentanal, pentan-1-ol

(2) n-butane, pentanal, ethoxyethane, pentan-1-ol

(3) pentanal, pentan-1-ol, ethoxyethane, n-butane

(4) pentanal, ethoxyethane, pentan-1-ol, n-butane

27. Among the following, which will react most rapidly with HBr?

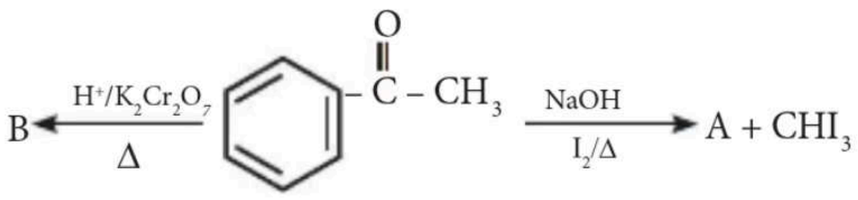
(1) $C_6H_5CH_2CH_2CH_2OH$

(2) $C_6H_5\underset{\displaystyle OH}{\underset{|}{C}}HCH_2CH_3$

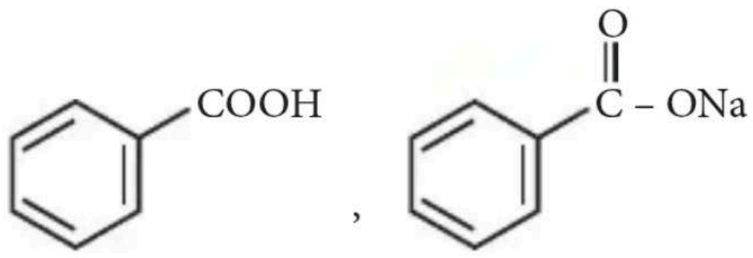
(3) $C_6H_5CH_2\underset{\displaystyle OH}{\underset{|}{C}}H - CH_3$

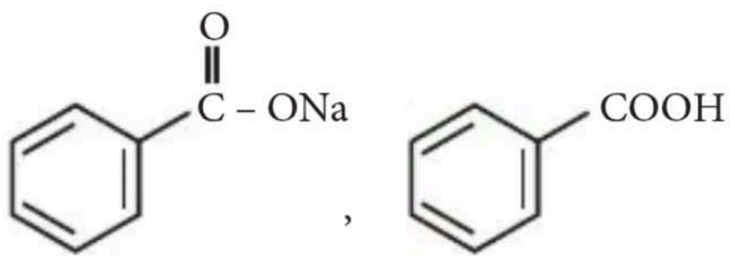
(4) $CH_3 - \underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}} - CH_2OH$

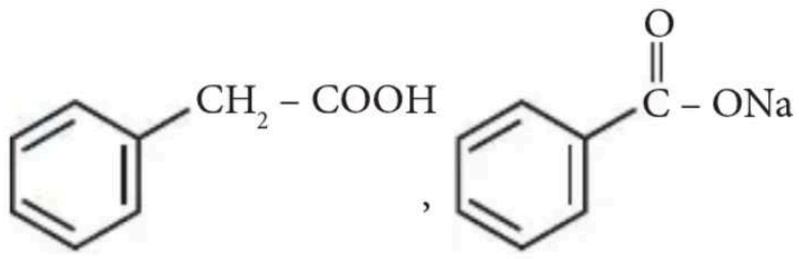
28. Read the following

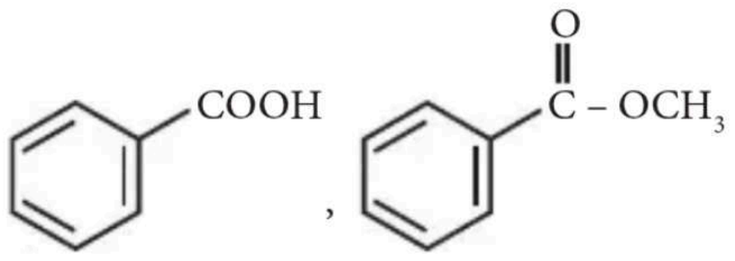


The products A and B respectively are

(1) 

(2) 

(3) 

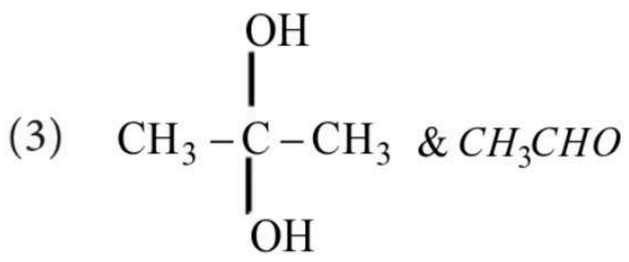
(4) 

29. Ethylidene chloride

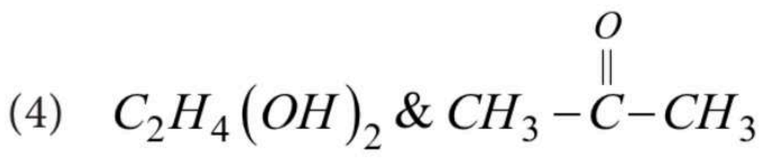
$\xrightarrow{2KOH(aq)} X \xrightarrow[\Delta]{-H_2O} Y.$ X and Y are

(1) C_2H_5OH and CH_3-CHO

(2) $C_2H_4(OH)_2$ and CH_3-CHO

(3) 

& CH_3CHO

(4) 

30. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Separation of Zr and Hf is difficult.

Reason (R) : Zr and Hf lie in the same group of the periodic table.

In light of the above statements, choose the correct answer from the options given below.

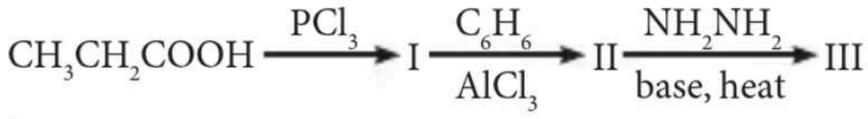
(1) Both A and R are correct and R is the correct explanation of A

(2) Both A and R are correct but R is not the correct explanation of A

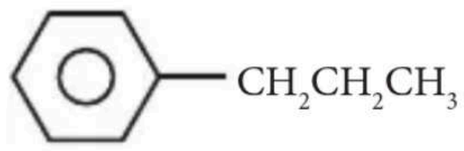
(3) A is correct but R is not correct

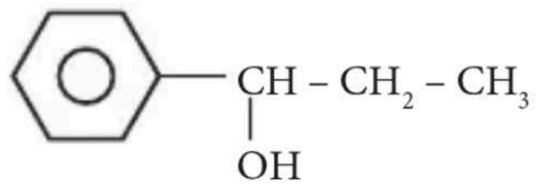
(4) A is not correct but R is correct

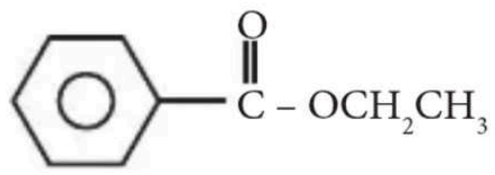
31. The final product (III) obtained in the reaction sequence

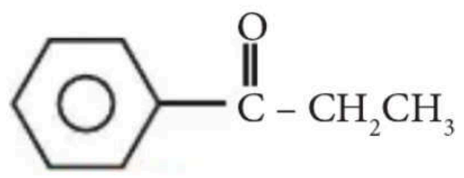


is

(1) 

(2) 

(3) 

(4) 

32. What type of amine is tertiary butylamine?

(1) Primary

(2) Secondary

(3) Tertiary

(4) Quaternary

33. Which reagent is used to distinguish between $(C_2H_5)_2NH$ and $(C_2H_5)_3N$?

(1) NaOH (2) Dilute HCl

(3) $C_6H_5SO_2Cl$ (4) $AgNO_{3(aq)}$

34. Which of the following indicates five OH groups in glucose?

(1) Penta acetyl derivative of glucose

(2) Cyanohydrin formation of glucose

(3) Reaction with Fehlings solution

(4) Reaction with Tollen's reagent

35. The solution of a chemical compound (X) reacts with $AgNO_3$ solution to form a white precipitate of 'Y' which dissolves in NH_4OH to give a complex *Z*. The chemical compound 'X' can be :

(1) NaCl (2) CH_3Cl

(3) NaBr (4) NaI

36. Given below are two statements:

Statement I : If ΔG is negative the reaction is spontaneous.

Statement II : If ΔG° is positive, the value of equilibrium constant is less than 1.

In light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

37. $K_2[HgI_4]$ detects the ion/group

(1) NH_2 (2) NO

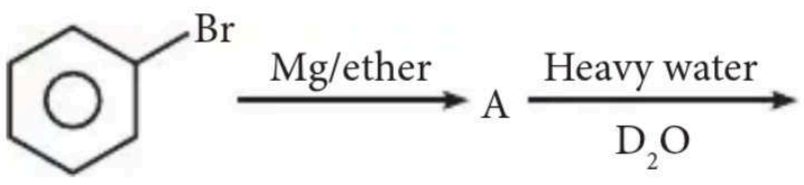
(3) NH_4^+ (4) Cl^-

38. $E^0_{M^{+2}/M}$ of which two elements of 3d series in transition elements have same value

(1) V, Mn (2) V, Cu

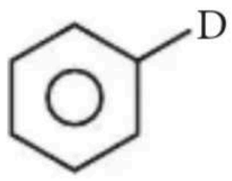
(3) Ti, Cr (4) Fe, Co

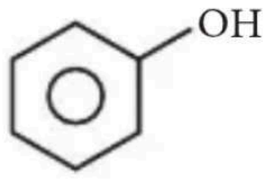
39. The final product obtained in the reaction

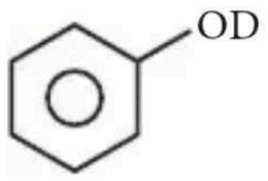


is

(1) 

(2) 

(3) 

(4) 

40. Which of the following statements is ***incorrect*** about f-block elements

(1) Many trivalent lanthanoid ions are coloured both in the solid state and in aqueous solutions.

(2) The first ionisation enthalpies of the lanthanoids are around 600 kJ mol^{-1}, the second about 1200 kJ mol^{-1} comparable with those of calcium.

(3) The actinoids resemble the lanthanoids in having more compounds in +3 state than in the +4 state.

(4) Thorium shows variable oxidation states of +4, + 5, + 6.

41. ***Correct*** statements among the following

A. XeO_3 is a colourless, explosive solid and has a pyramidal molecular structure

B. $XeOF_4$ is a colourless volatile liquid and has a square pyramidal molecular structure

C. Xenon fluorides react with fluoride ion acceptors to form cationic species and with fluoride ion donors they form anionic species

D. Ne is used in the laboratory for handling substances that are air sensitive

(1) A, B and C only (2) B and C only

(3) A and C only (4) A, B, C, D

42. Read the following

A. $I^- + S_2O_3^{2-} \rightarrow I_2 + S_4O_6^{2-}$. In this reaction, n-factor of thiosulphate ion is one.

B. In Pb_3O_4, the oxidation states of each oxygen is –2 and –1 respectively.

C. In the reaction of oxalic acid and potassium permanganate in acidic medium, the total number of electrons involved in the net reaction is ten.

D. Thermal decomposition of $KMnO_4$, $KClO_3$, $NaNO_3$ are all examples of disproportionation reactions.

Correct statements are

(1) B and C only (2) A, B, C, D

(3) A, B and C (4) A and C only

43. When n-hexane is heated with Mo_2O_3 at 500°C and 20 atm, changes taking place are

A. Hybridisation of carbon changes

B. Number of C-C bonds increase

C. Number of C-H bonds decrease

D. Percentage of Carbon increases

(1) A, B, C and D (2) A, C and D

(3) A, B and C (4) B, C and D

44. Match the entries of List-I with appropriate entries of List-II and select the correct answer using the codes given below the lists.

	List-I	List-II
P.	Rutherfordium (At. No. = 104)	1. Period number = 7
Q.	Roentgenium (At. No. = 111)	2. Group number = 4
R.	Thorium (At. No. = 90)	3. d-block elements
S.	Neptunium (At. No. = 93)	4. f-block elements

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	3	2,4	2,3	1,2,3
(2)	1,2,3	1,3	1,4	1,4
(3)	1,2	2,1	4,3	3,1
(4)	3,1	1,3	2,4	4,1,2

45. Given below are two statements :

Statement-I : Resultant dipole moment of NH_3 molecule is greater than that of NF_3 molecule.

Statement-II: CO_2 is a non-polar molecule with polar bonds.

In light of the above statements, choose the correct answer from the options given below.

(1) Both Statement I and Statement II are true

(2) Statement I is false but Statement II is true

(3) Statement I is true but Statement II is false

(4) Both Statement I and Statement II are false

ZOOLOGY

46. Which of the following are the two main components of a meromyosin molecule?

(1) A globular head with calcium binding sites and a tail

(2) Heavy meromyosin and the light meromyosin

(3) Cross arm and F actin

(4) Troponin and tropomyosin

47. Which of the following animals exhibit segmentation?

(1) *Sycon* and *Aurelia*

(2) *Pinctada* and *Obelia*

(3) *Pheretima* and *Periplaneta*

(4) *Ancylostoma* and *Balanoglossus*

48. Which of the following statements are ***correct*** about rib cage?

A. Thoracic vertebrae, ribs and sternum together forms rib cage.

B. Floating ribs do not articulate with sternum.

C. The 8th, 9th and 10th pairs of ribs are vertebrosternal ribs.

D. First seven pairs of ribs are called true ribs.

(1) A, B, and D are correct

(2) A, B, and C are correct

(3) Only B and C are correct

(4) B, C, and D are correct

49. Read the following statements and select ***correct*** options.

A. Larval forms of echinoderms exhibit bilateral symmetry.

B. Adult cnidarians exhibit bilateral symmetry.

C. Majority of sponges do not exhibit symmetry.

(1) Only A is correct

(2) Only B is correct

(3) A and C are correct

(4) Only C is incorrect

50. Given below are two statements.

Statement I : During inspiration diaphragm is relaxed and arched to increase thoracic volume.

Statement II : Contraction of external inter-costal muscles increases the volume of the thoracic chamber in the dorso-ventral axis.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are true.

(2) Statement I is false but statement II is true.

(3) Statement I is true but statement II is false.

(4) Both statement I and statement II are false.

51. Match the following.

	List–I		List–II
A.	Metagenesis	**I.**	Annelida
B.	Parapodia	**II.**	Aschelminthes
C.	Pseudocoelom	**III.**	Platyhelminthes
D.	Triploblastic acoelomates	**IV.**	Cnidarians

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	IV	I	II	III
(2)	I	IV	II	III
(3)	III	II	IV	I
(4)	III	IV	I	II

52. Which of the following is a special venous system present in between the gut and the liver of frog?

(1) Hepatic portal system

(2) Hypophyseal portal system

(3) Renal portal system

(4) Coronary circulatory system

53. Select the ***correct*** statements from the following.

I. In human beings, after one month of pregnancy, the embryo's heart is formed.

II. By the end of the second month of pregnancy, the foetus develops limbs and digits.

III. By the end of the first month, the limbs and external genital organs are well developed.

IV. The first movements of the foetus usually observed during the fifth month.

(1) I, III, and IV

(2) I and II

(3) I and III

(4) I, II, and IV

54. The mouth parts of cockroach include all, except

(1) mandibles and maxillae

(2) labrum and labium

(3) mandibles and hypopharynx

(4) tegmina and phallomeres

55 Match the following.

	List I		List II
A.	Dense irregular connective tissue	**I.**	Ear pinna
B.	Elastic cartilage	**II.**	Bowman's capsule
C.	Squamous epithelium	**III.**	Dermis of skin
D.	Columnar epithelium	**IV.**	Gastrointestinal tract

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	I	III	IV	II
(2)	III	I	II	IV
(3)	I	IV	II	III
(4)	I	II	IV	III

56. Which of the following are common to frogs and cockroaches?

A. Closed circulatory system

B. Cutaneous respiration

C. Unisexual nature

D. Omnivorous habit

(1) A and B are correct

(2) B and C are correct

(3) A and D are correct

(4) Only C is correct

57. Which of the following is true about the pneumotaxic centre?

(1) It can moderate the function of respiratory rhythm centre.

(2) This can recognise changes in CO_2 and Ca^{++} ion concentration in blood.

(3) It is located in the medulla oblongata.

(4) Strong neural signals from this can increase the duration of inspiration.

58. Inflammation of glomerular blood capillaries in the kidney is termed

(1) kidney failure

(2) glomerulonephritis

(3) diabetes mellitus

(4) diabetes insipidus

59. Given below are two statements.

Statement I : An excessive loss of fluid from the body can activate osmoreceptors which stimulate the hypothalamus to release vasopressin.

Statement II : Aldosterone causes reabsorption of Na^+ and water from the distal parts of the tubule.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are incorrect.

(2) Both statement I and statement II are correct.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct.

60. Which of the following is a hypothalamic hormone correctly matched with its function?

(1) TSH – Stimulates secretion of thyroxine

(2) PRL – Suppresses the secretion of prolactin

(3) GnRH – Stimulates the secretion of gonadotropins

(4) ACTH – Stimulates the release of adrenal medullary hormones

61. Corpora quadrigemina are

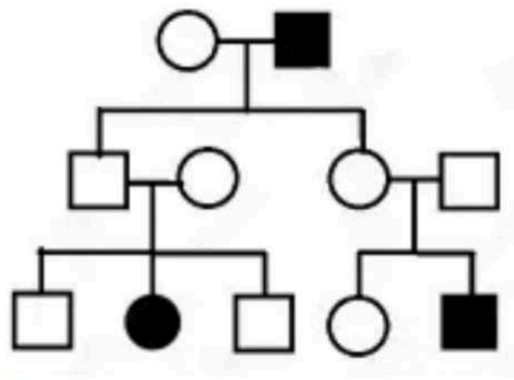
(1) four rounded swellings on the dorsal portion of the midbrain.

(2) nerve fibre tracts that connect two cerebral hemispheres

(3) ventricles of the midbrain filled with cerebrospinal fluid

(4) four rounded swellings at dorsal portion of the cerebellum

62. In the given pedigree chart, the gene responsible for the disorder is



(1) sex linked dominant

(2) sex linked recessive

(3) autosomal dominant

(4) autosomal recessive

63. Select the ***correct*** combination.

	Blood group	Antigens on RBCs	Antibodies in Plasma
(1)	A	B	anti-B
(2)	B	A	anti-A
(3)	AB	nil	anti-A, anti-B
(4)	O	nil	anti-A, anti B

64. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : An impairment affecting synthesis or release of ADH results in diabetes insipidus.

Reason (R) : Posterior pituitary stores and releases ADH.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

65. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : The nodal musculature has the ability to generate action potentials without any external stimuli.

Reason (R) : Both SAN and AVN are associated with the right atrium.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

66. Which of the following statements is ***incorrect*** about human beings?

(1) Fertilisation occurs in the fallopian tube.

(2) The secondary oocyte forms a new multicellular membrane called the zona pellucida.

(3) The blastocyst becomes embedded in the endometrium, leading to pregnancy.

(4) Both first and second meiotic divisions in oogenesis are unequal.

67. In the absence of fertilisation, corpus luteum

(1) secretes FSH and LH

(2) produces oxytocin and relaxin

(3) degenerates leading to menstruation

(4) causes amenorrhoea

68. Given below are two statements.

Statement A : Periodic abstinence is a natural method in which couples avoid coitus from day 10 to 17 of the menstrual cycle.

Statement B : Lactational amenorrhea is a permanent natural contraceptive method.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement A and statement B are false.

(2) Statement A is true, statement B is false.

(3) Both statement A and statement B are true.

(4) Statement A is false, statement B is true.

69. Match the following.

	List I		List II
A.	Secretory phase	**I.**	Maturation of follicle
B.	Proliferative phase	**II.**	Development of corpus luteum
C.	Menstruation	**III.**	Breakdown of endometrial lining
D.	Ovulation	**IV.**	LH surge

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	III	II	IV	I
(2)	I	IV	II	III
(3)	II	I	III	IV
(4)	III	IV	I	II

70. ELISA is based on the principle of

(1) antigen-antibody interaction

(2) *r*DNA technology

(3) polymerase chain reaction

(4) DNA polymorphism

71. Select the ***incorrect*** match of the an assisted reproductive technology technique and its feature.

(1)	IVF	Fertilisation outside the body in almost similar conditions to those in the body
(2)	ICSI	Injection of sperm directly into the ovum
(3)	AI	Introducing semen into the vagina or the uterus
(4)	GIFT	Transfer of sperm but not ovum into the fallopian tube of a recipient

72. In an ideal energy pyramid, primary producers convert only _____ of the energy present in the sunlight available to them into NPP.

(1) 1% (2) 20-25%

(3) 10% (4) 50%

73. Which of the following is a hormonal IUD?

(1) Multiload-375

(2) LNG-20

(3) CuT

(4) Lippes loop

74. Which of the following sets is ***not*** an example of divergent evolution?

(1) Forelimbs of man, bat and cheetah

(2) Hearts of bat, man and cheetah

(3) Brains of bat, man and cheetah

(4) Eyes of octopus and of a man

75. Select the ***correct*** statements.

I. Haemophilia is a sex-linked recessive disease.

II. Down's syndrome is due to aneuploidy.

III. Phenylketonuria is an autosomal recessive gene disorder.

IV. Sickle cell anaemia is an X-linked recessive gene disorder.

(1) II and IV are correct

(2) I, III and IV are correct

(3) I, II and III are correct

(4) I and IV are correct

76. Synthesis of alpha globin chains of haemoglobin molecules is controlled by

(1) a gene located on 21st chromosome

(2) HBA1 and HBA2 linked genes located on the X-chromosome

(3) two closely linked genes located on the 16th chromosome

(4) two closely linked genes located on the 11th chromosome

77. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Micro-satellites and mini-satellites contain sequences that normally do not code for any protein.

Reason (R) : Satellite DNA is mainly composed of Expressed Sequence Tags.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) (A) is false but (R) is true.

78. Select the ***incorrect*** statements from the following regarding organic evolution.

I. About 65 mya, the dinosaurs suddenly disappeared from the Earth.

II. Due to continental drift, pouched mammals of South Africa survived because of lack of competition.

III. Sea weeds and a few plants existed probably around 320 mya.

IV. Several million years ago, bryophytes were present, but they fell to form coal deposits slowly.

(1) I, III, and IV

(2) I and II

(3) I and III

(4) II and IV

79. Which of the following is the most important cause of animals and plants being driven to extinction?

(1) Co-extinction

(2) Overexploitation

(3) Alien species invasion

(4) Habitat loss and fragmentation

80. Cannabinoids are known for their effects on

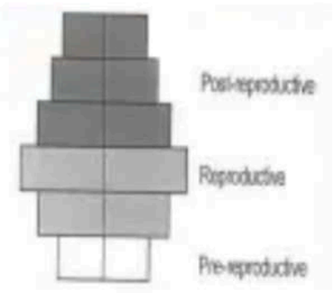
(1) digestive system

(2) respiratory system

(3) excretory system

(4) cardiovascular system

81. What type of human population is represented by the following age pyramid?



(1) Expanding population

(2) Flourishing population

(3) Stable population

(4) Declining population

82. Sexual stages of *Plasmodium* develop in

(1) red blood cells of mosquito

(2) red blood cells of man

(3) small intestine of man

(4) salivary glands of mosquito

83. Given below are two statements.

Statement I : The standing crop is measured as the mass of living organisms or the number in a unit area.

Statement II : The base of all three types of ecological pyramids represents the producers.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are incorrect.

(2) Both statement I and statement II are true.

(3) statement I is correct, statement II is incorrect.

(4) Statement I is incorrect, statement II is correct.

84. Given below are two statements.

Statement I : Benign tumors normally remain confined to their original location and do not cause any damage.

Statement II : Metastasis is the most feared property of malignant tumors.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Statement I is correct, statement II is incorrect.

(3) Statement I is incorrect, statement II is correct.

(4) Both statement I and statement II are incorrect.

85. Match the following.

	List-I		List-II
A.	A & B chains of insulin	**I.**	Used in gene therapy
B.	Functional ADA cDNA	**II.**	PCR
C.	Amplification of DNA	**III.**	Held by disulphide bridges
D.	Alpha-1 antitrypsin	**IV.**	Helps to treat emphysema

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	IV	I	II	III
(2)	III	II	IV	I
(3)	IV	I	III	II
(4)	III	I	II	IV

86. Select the ***correct*** statements about predation.

A. Predators help maintain species iversity in a community by reducing the intensity of competition among competing prey species.

B. In predation, one species benefits while the other is neither benefited nor harmed.

C. Predators keep prey populations under control.

D. Predators act as conduits for energy transfer across trophic levels.

(1) A, C, and D only

(2) A and C only

(3) A and B only

(4) Only B is correct

87. Among the following, who was the first to use animal hides to protect their body?

(1) Java man

(2) Peking man

(3) Neanderthal man

(4) Modern man

88. Select the ***correct*** statements about Indian biological diversity.

A. India has only 2.4 per cent of the world's land area.

B. India ranks first in the world in terms of biodiversity.

C. Indian share of the global species diversity is 8.1 per cent.

D. India hosts more than 1,200 species of birds.

E. Out of 34 biodiversity hot spots, seven are present in India.

(1) A, B, and D (2) A, B, and C

(3) A, C, and D (4) B, D, and E

89. Which of the following is the key pollinator of fig trees?

(1) Female bumble bee

(2) Male bumble bee

(3) Female wasp

(4) Male wasp

90. Select the ***incorrect*** match.

(1) Steller's Sea Cow – Russia

(2) Dodo – Mauritius

(3) Quagga – Africa

(4) *Thylacine* – America

PHYSICS

91. Three bodies A, B and C are having temperature $T_A = 38^o F$, $T_B = -5^o C$, $T_C = 200$ K respectively. Which body among these is most warm?

(1) A

(2) B

(3) C

(4) All are equally warm

92. A liquid does ***not*** wet solid surface if angle of contact is

(1) 20° (2) 70°

(3) 110° (4) 88°

93. A black body has maximum intensity wavelength λ_m at 500 K. The wavelength of maximum intensity at 3000 K is

(1) $3\lambda_m$ (2) $\frac{\lambda_m}{3}$

(3) $6\lambda_m$ (4) $\frac{\lambda_m}{6}$

94. If water is heated from $0^o C$ to $10^o C$, then the volume

(1) Increase throughout

(2) decrease throughout

(3) Remains constant

(4) First decreases then increases

95. Two wires made of the same material have masses in the ratio 3:2 and lengths in the ratio 1:2. If these wires are stretched by loads in the ratio 3:4, the ratio of strain produced in the two wires is

(1) 1 : 4 (2) 9 : 16

(3) 3 : 32 (4) 9 : 32

96. If the kinetic energy of an orbiting artificial satellite is K_o, then the potential energy and binding energy of satellite are

(1) $2K_o, K_o$

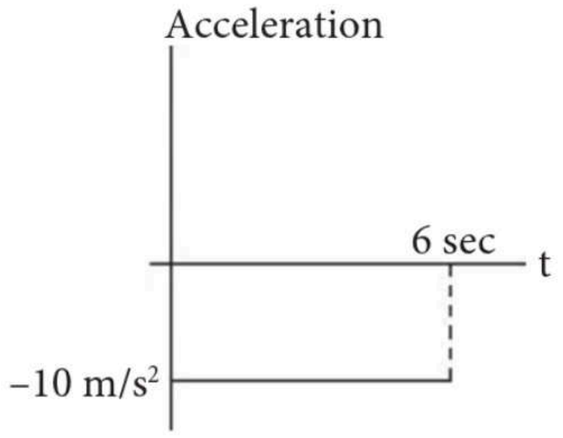
(2) $-2K_o, -K_o$

(3) $-2K_o, K_o$

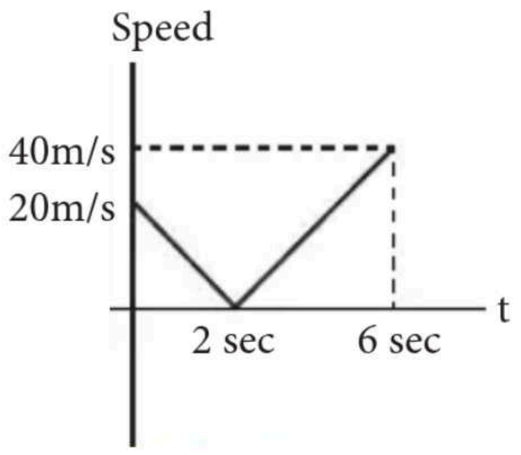
(4) K_o, K_o

97. A stone is thrown vertically upward from the top of a tower of height 60 m with initial speed of 20 m/s. Which of following graph (s) is (are) possible. (Take ground as origin and upward direction as positive) ($g = 10$ m/s^2)

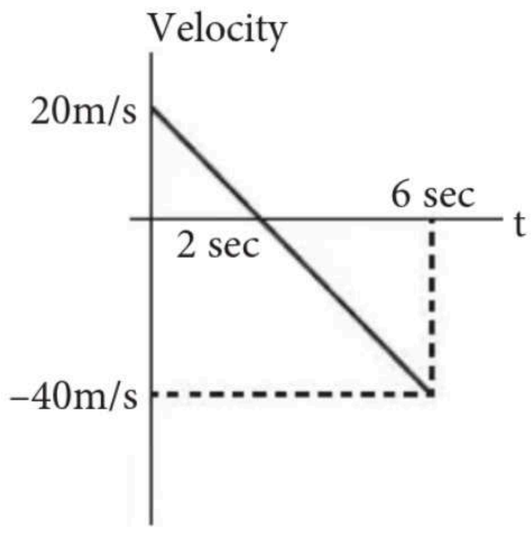
(i)



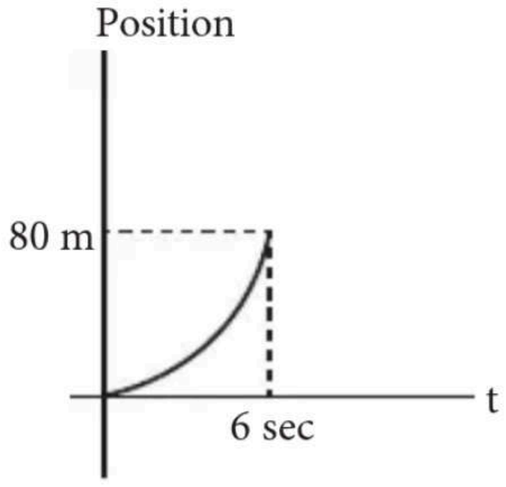
(ii)



(iii)



(iv)



(1) (i) ,(ii), and (iv) only

(2) (i) and (iv) only

(3) (i), (ii), (iii), and (iv)

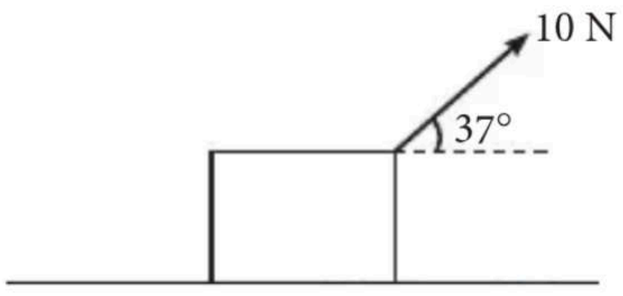
(4) (i), (ii), and (iii) only

98. A stone is thrown vertically upward with an initial speed of V_0. At a certain point (P) its kinetic energy is 4 times of its potential energy. Calculate height of point P from ground.

(1) $\frac{V_0^2}{2g}$ (2) $\frac{V_0^2}{5g}$

(3) $\frac{V_0^2}{10g}$ (4) $\frac{V_0^2}{g}$

99. A body of mass 5 kg is at rest on a horizontal table. The coefficient of friction between the body and the table is 0.6. A force of magnitude 10 N is acting on the block at an angle 37° with the horizontal then friction force acting between the block and the table will be :



(1) 30 N

(2) 26.4 N

(3) 13.2 N

(4) 8 N

100. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Silicon is preferred over germanium for making semiconductor devices.

Reason (R) : The energy band for germanium is more than the energy band of silicon.

In light of the above statements, choose the correct answer from the options given below.

(1) Both 'A' and 'R' are true and 'R' is the correct explanation of 'A'.

(2) Both 'A' and 'R' are true and 'R' is not the correct explanation of 'A'

(3) 'A' is true and 'R' is false

(4) Both 'A' and 'R' are false

101. Rain is falling vertically with a speed 20 m/s. A girl rides a bicycle with a speed of 15 m/s due west. In which direction she should hold her umbrella?

(1) 53° with vertical towards west

(2) 37° with vertical towards west

(3) 37° with vertical towards east

(4) 53° with vertical towards east

102. Given below are two statements.

Statement 1 : Static friction is self adjusting force upto its limit $\mu_s N$ ($f_{static} \leq \mu_s N$)

Statement 2 : Total momentum of an isolated system of particle is conserved. This law follows from second law of motion.

(1) Both statements are correct

(2) Both statements are incorrect

(3) Statement 1 is correct and statement 2 is incorrect

(4) Statement 1 is incorrect and statement 2 is correct

103. Choose ***incorrect*** statements

A. In an elastic collision of two bodies, the momentum and energy of each body is conserved.

B. In an elastic collision, the final kinetic energy is always less than the initial kinetic energy of the system.

C. Work done in the motion of a body over a closed loop is zero for every force in nature.

D. Work done by the resistive force of air on a vibrating pendulum in bringing it to rest is positive.

(1) A, B, and C only

(2) A, B, C, and D

(3) B, C, and D only

(4) B and D only

104. A body of mass 1 kg travels in a straight line with velocity $v = 5x^{3/2}$ m/s. The work done by net force during its displacement from x = 0 to x = 2m is

(1) 50 J (2) 100 J

(3) 150 J (4) $200\sqrt{3}$ J

105. During an elastic collision

Statement A : The total linear momentum is conserved at each instant of the collision.

Statement B : The kinetic energy conservation applies after the collision is over and does not hold at every instant of the collision.

Statement C : When two equal masses undergo a glancing elastic collision with one of them at rest then after the collision, they will move at right angles to each other.

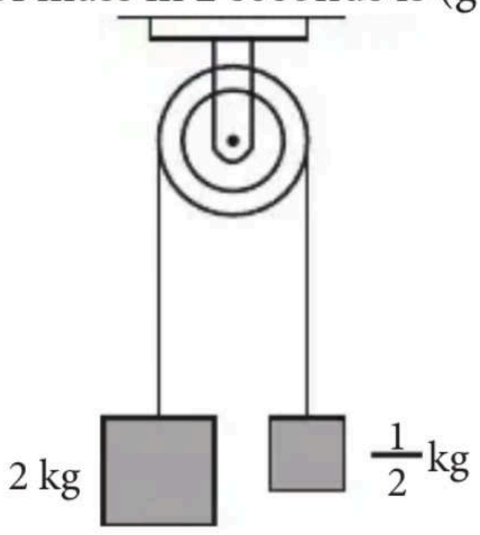
(1) A and B are true

(2) A and C are true

(3) B and C are true

(4) A, B and C are true

106. Two blocks of masses 2 kg and $\frac{1}{2}$ kg respectively are tied to the ends of a string which passes over a light frictionless pulley. The masses are held at rest at the same horizontal level and then released. The distance transverse by centre of mass in 2 seconds is ($g = 10\ m/s^2$)



(1) 2.2 m (2) 7.2 m

(3) 5.4 m (4) 6.6 m

107. Two equal masses '*m*' are revolving in a circleof radius *r* due to mutual interaction (gravitational force). Calculate linear speed of each mass.

(1) $\sqrt{\frac{GM}{r}}$ (2) $\frac{1}{4}\sqrt{\frac{GM}{r}}$

(3) $\sqrt{\frac{GM}{4r}}$ (4) $\sqrt{\frac{GM}{2r}}$

108. The dimension of $\mu_0 \in_0$ is

(1) LT^{-1}

(2) $L^{-2}T^{-2}$

(3) $L^2 T^2$

(4) $L^{-2}T^2$

109. In an electric circut, the device that represents electrical inetria is

(1) Inductor

(2) Capacitor

(3) Resistor

(4) Battery

110. If electric potential is $V = \left(4x^2y + 2yz^2 + 10x\right)$ volts. The electric field at P(1, 2, 3)m is:

(1) $\left(13\hat{i} + 11\hat{j} + 12\hat{k}\right) V/m$

(2) $-\left(13\hat{i} + 11\hat{j} + 12\hat{k}\right) V/m$

(3) $-\left(26\hat{i} + 22\hat{j} + 24\hat{k}\right) V/m$

(4) $\left(26\hat{i} + 22\hat{j} + 24\hat{k}\right) V/m$

111. A light of wavelenrth 400 nm is incident on the photo – sensitive metal. If stopping potential is 1V, calculate threshold wavelength

(1) 620 nm (2) 590 nm

(3) 450 nm (4) 650 nm

112. Saturation photo current depends on (in photoelectric conditions)

(1) Frequency of light

(2) Energy of photon

(3) Polishing of metal surface

(4) Intensity of light

113. Bernoulli's principle is based upon

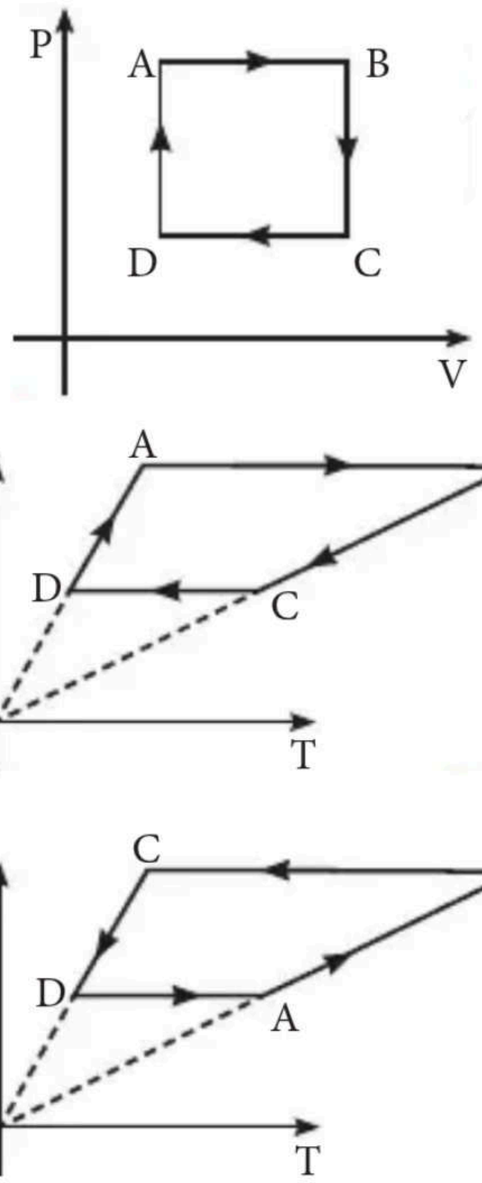
(1) Pascal's law

(2) Archimede's Principle

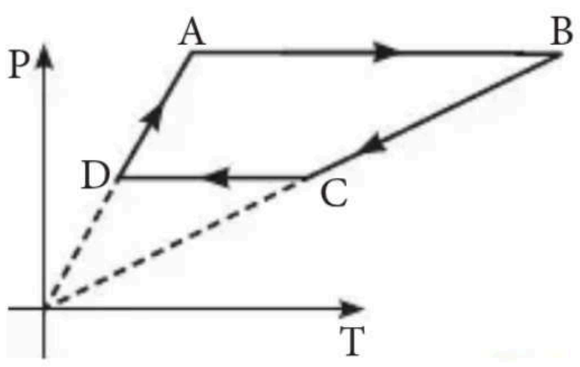
(3) Law of conservation of linear momentum

(4) Law of conservation of energy

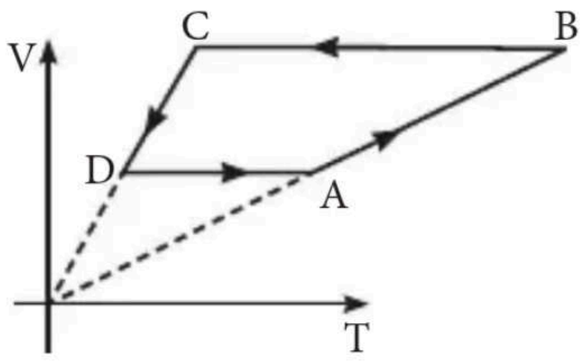
114. Which of following options is ***correct*** for the given graph?



(1)



(2)



(3) Both graphs (1) & (2) are correct

(4) Both graphs (1) & (2) are not correct

115. If internal energy of a gas is given by U = (3 PV + 4). The degrees of freedom of gas are

(1) 3 (2) 4

(3) 5 (4) 6

116. A physical quantity X is given by $X=\frac{2k^3l^2}{m\sqrt{n}}$. The percentage error in the measurements of k, l, m and n are 1%, 2%, 3% and 4% respectively. The value of X is uncertain by

(1) 8% (2) 10%

(3) 12% (4) 6%

117. An electromagnetic radiation is travelling in positive Z–direction. Which of the following can be direction of electric field and magnetic field respectively?

(1) $\left(\hat{j}+\hat{k}\right),\left(\hat{j}-\hat{k}\right)$ (2) $\left(\hat{j}+\hat{k}\right),\left(-\hat{j}+\hat{k}\right)$

(3) $\left(\hat{i}+\hat{j}\right),\left(\hat{i}-\hat{j}\right)$ (4) $\left(\hat{i}-\hat{j}\right),\left(\hat{i}+\hat{j}\right)$

118. The dimensional formula of Poynting vector $\vec{S}=\frac{1}{\mu_0}\left(\vec{E}\times\vec{B}\right)$ is

(here $\vec{E}$ is electric field, $\vec{B}$ is magnetic field, μ_o is permeability of free space)

(1) $\left[ML^2T^{-2}\right]$

(2) $\left[ML^2T^{-3}\right]$

(3) $\left[ML^0T^{-3}\right]$

(4) $\left[ML^0T^{-2}\right]$

119. In a double slit experiment, when light of wavelength 800 nm was used, the angular width of the first minima formed on a screen placed 1m away, was found to be 0.4^0. What will be the angular width of the first minima, if the entire experimental apparatus is immersed in water? $\left(\mu_{water}=4/3\right)$:

(1) 0.266^0 (2) 0.15^0

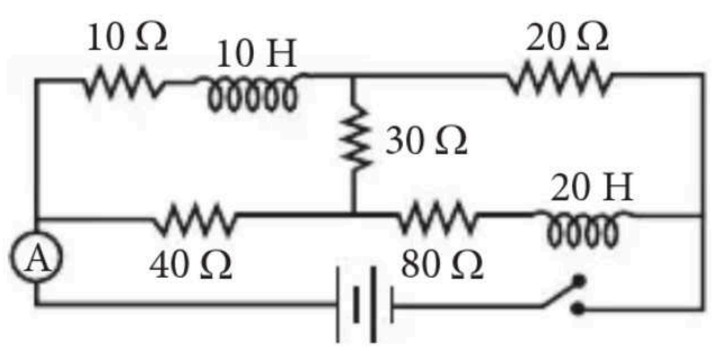
(3) 0.05^0 (4) 0.3^0

120. An ac voltage is applied to a resistance R and an inductor L in series. If R and the inductive reactance each equal to 3Ω, the phase difference between the applied voltage and the current in the circuit is

(1) π/6 (2) π/4

(3) π/2 (4) zero

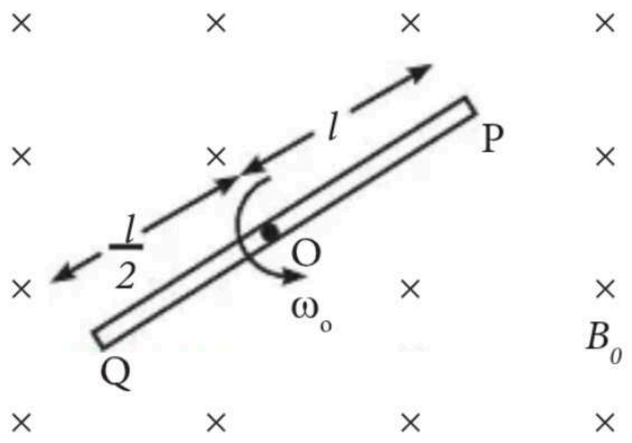
121. Calculate the reading of ammeter when switch is just closed



(1) 20 A (2) 0.2 A

(3) 500 mA (4) 5 A

122. The magnitude of potential potential difference between P and Q is



(1) $\frac{3}{8}B_o\omega_o l^2$ (2) $\frac{3}{4}B_o\omega_o l^2$

(3) $\frac{1}{8}B_o\omega_o l^2$ (4) $\frac{1}{4}B_o\omega_o l^2$

123. Self-inductance of a solenoid depends on

(1) Number of turns of the solenoid

(2) Radius of solenoid

(3) Permeability of core of solenoid

(4) All of the above

124. Above the Curie's temperature

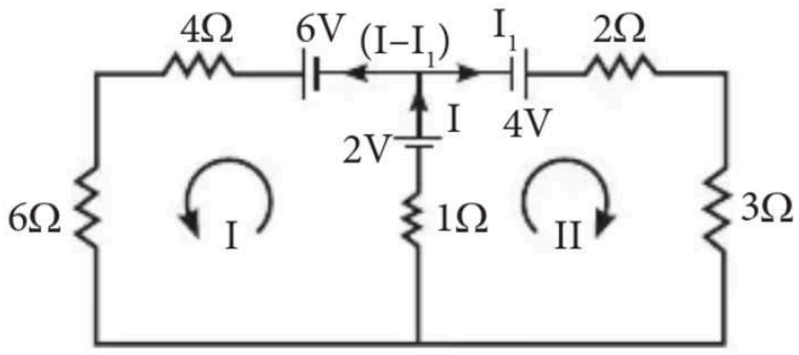
(1) Non–radioactive substance converts into radioactive

(2) Ferromagnetic substance converts into diamagnetic

(3) Radio active substance converts into non–radioactive

(4) Ferromagnetic substance converts into paramagnetic

125. Which of the following Kirchhoff's loop equation is ***correct*** for loop I?



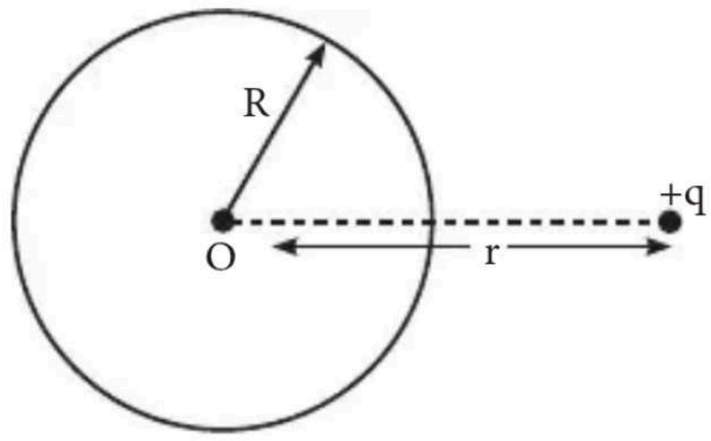
(1) $6V+4(I-I_1)+6(I-I_1)-1(I)+2V=0$

(2) $-6V+4(I-I_1)+6(I-I_1)+1(I)+2V=0$

(3) $-6V+4(I-I_1)+6(I-I_1)+1(I)-2V=0$

(4) $6V-4(I-I_1)-6(I-I_1)+1(I)+2V=0$

126. The magnitude of electric field and electric potential at O due to induced charge on metallic sphere respectively are

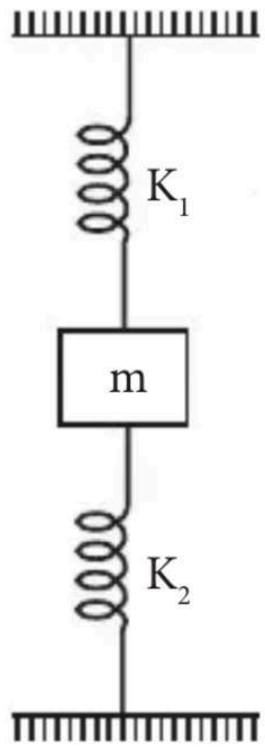


Metallic sphere

(1) 0, 0 (2) $0, \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}$

(3) $\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}, \frac{1}{4\pi\varepsilon_0}\frac{q}{r}$ (4) $\frac{1}{4\pi\varepsilon_0}\frac{q}{r^2}, 0$

127. Assuming ideal springs, the frequency of oscillation of block is



(1) $\frac{1}{2\pi}\sqrt{\frac{m}{(K_1+K_2)}}$

(2) $\frac{1}{2\pi}\sqrt{\frac{K_1+K_2}{m}}$

(3) $\frac{1}{2\pi}\sqrt{\frac{m}{\left(\frac{1}{K_1}+\frac{1}{K_2}\right)}}$

(4) $\frac{1}{2\pi}\sqrt{m\left(\frac{1}{K_1}+\frac{1}{K_2}\right)}$

128. If 3rd overtone of open end organ pipe of length 20cm resonates with 2nd overtone of closed end organ pipe of length l, then value of '$4l$' is

(1) 12.5 cm

(2) 25 cm

(3) 100 cm

(4) 50 cm

129. 2 mole of Helium and 4 moles of Oxygen gas is mixed in a closed container and maintained at temperature 'T' Kelvin. The internal energy of the mixture is

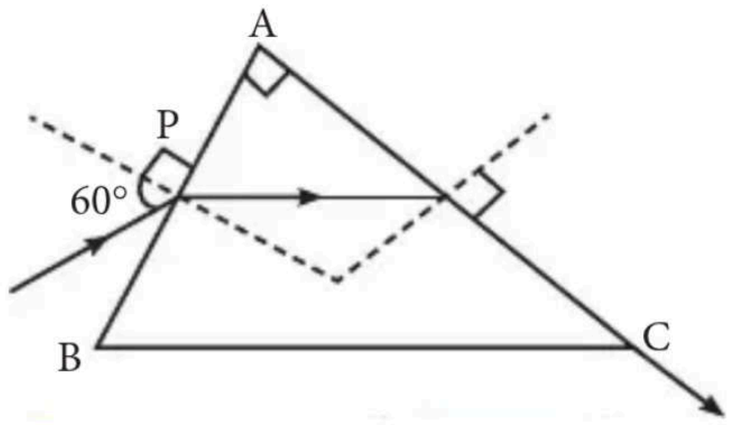
(1) 13 RT

(2) 26 RT

(3) 6 RT

(4) 10 RT

130. A light ray enters through a right angled prism at point P with the angle of incidence 60° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is:



(1) $\frac{\sqrt{5}}{2}$

(2) $\frac{\sqrt{7}}{2}$

(3) $\frac{\sqrt{3}}{2}$

(4) $\frac{\sqrt{5}}{4}$

131. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Gauss's law cannot be used to calculate electric field due to an electric dipole.

Reason (R) : Electric dipole have spherical symmetrical charge distribution.

In light of the above statements, choose the correct answer from the options given below.

(1) Both A are R are true, R is the correct explanation of A

(2) Both A are R are true, R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

132. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Thermonuclear fusion reactions may become the source of unlimited power for the mankind.

Reason (R) : A single fusion event involving isotopes of hydrogen produces more energy than energy from nuclear fission of $^{235}_{92}U$.

In light of the above statements, choose the correct answer from the options given below.

(1) Both A are R are true, R is the correct explanation of A

(2) Both A are R are true, R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

133. A nucleus of mass number 220 decays by α decay. The energy released in the reaction is 5 MeV. The kinetic energy of an α–particle is

(1) $\frac{1}{54} MeV$

(2) $\frac{27}{11} MeV$

(3) $\frac{54}{11} MeV$

(4) $\frac{55}{54} MeV$

134. The nucleus ${}^{m}_{n}X$ emits one α – particle and $2\beta^-$ –particles. The resulting nucleus is

(1) ${}^{m-4}_{n-2}Y$

(2) ${}^{m-6}_{n}Z$

(3) ${}^{m-6}_{n}Z$

(4) ${}^{m-4}_{n}X$

135. In a transformer, the source of energy loss and the precaution to minimize that are mentioned in set-I and in set –II. Match the appropriate ones

	Set-I		Set-II
A.	Flux leakage	P.	Thin sheets of laminated core
B.	Resistance of windings	Q.	Thick wire in high current windings
C.	Eddy currents	R.	Soft iron core .
D.	Hysteresis	S.	Primary and secondary coils are wound one over the other

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	S	Q	R	P
(2)	S	Q	P	R
(3)	Q	P	S	R
(4)	S	R	P	Q

BOTANY

136. Read the following statements and pick the ***correct*** pair.

A. Ovules generally differentiate a single megaspore mother cell in the micropylar region of the nucellus.

B. In *Citrus* and mango varieties, some of the nucellar cells protrude into embryo sac and develop into apomictic embryos.

C. In seeds of black pepper and beet, residual persistent nucellus named endosperm is present.

D. Cells of nucellus possess dense cytoplasm with more than one nucleus and surround microsporangium completely.

(1) A and B.

(2) C and D.

(3) A and D.

(4) B and D.

137. Given below are two statements.

Statement I : An amino acid namely glutamic acid is replaced by valine in a polypeptide chain due to a point mutation resulting in sickle cell anaemia.

Statement II : An amino acid phenylalanine is replaced by tyrosine in a polypeptide chain due to a single gene mutation resulting in Phenylketonuria.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

138. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Ethylene helps to enhance yield in cucumber.

Reason (R) : Usage of ethylene promotes female flowers in cucumber.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

139. Read the following statements and pick the ***incorrect*** one.

(1) RER is observed in cells involved in protein synthesis and secretion.

(2) Gas vacuole in bacteria is important for osmoregulation and secretion.

(3) Golgi complex helps in packaging materials to be delivered either to intracellular targets (or) secreted out of the cell.

(4) SER is a major site for synthesis of lipids.

140. Pick the ***incorrect*** match.

(1) Carotenoid derivative : ABA

(2) Adenine derivative : Cytokinins

(3) Terpene derivative : GA

(4) Indole derivative : Ethylene

141. Select the ***correct*** option by placing the following in ***correct*** order of their usage in sewage treatment plant.

A. Secondary settling tank

B. Aeration tank

C. Anaerobic sludge digestors

D. Primary settling tank

(1) A B C D

(2) D A B C

(3) D B A C

(4) D B C A

142. Read the following statements and select the ***incorrect*** one.

(1) Generally roots are positively geotropic, but in *Rhizophora* they are negatively geotropic.

(2) Generally flowers in dicots have pentamerous symmetry, but in *Brassica* they are trimerous.

(3) Generally leaves perform photosynthesis, but in *Opuntia* stem performs the photosynthesis.

(4) Generally dicot seeds are non-endospermic, but *Ricinus* (castor) seeds are endospermic.

143. Who demonstrated that photosynthesis is light driven process in which hydrogen from suitable oxidisable compounds reduces carbon dioxide to carbohydrates?

(1) T.W. Engelmann

(2) C. van Neil

(3) Jan Ingenhousz

(4) Joseph Priestly

144. Pick the ***incorrect*** match from the following.

(1) Pappus : hairy sepals in Sunflower.

(2) Epicalyx : bracteoles in China rose.

(3) Keel : largest posterior petal in pea.

(4) Lodicules : reduced perianth in grasses.

145. Given below are two statements. One is labelled Assertion (A) and the other is labelled Reason (R).

Assertion (A) : Cleistogamous flowers are invariably self - pollinated (Autogamy).

Reason (R) : In cleistogamous flowers, there is no chance of foreign pollen landing on stigma as they do not open.

In light of the above statements, choose the correct answer from the options given below.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(3) (A) is true, but (R) is false.

(4) Both (A) and (R) are false.

146. A 3C organic acid named phosphoglyceric acid (PGA) is produced in all the listed metabolic pathways except one. What is it?

(1) Calvin cycle.

(2) Tricarboxylic acid cycle.

(3) Hatch- Slack pathway.

(4) Glycolysis.

147. 5 - methyl uracil (thymine) is observed in

(1) *hn*RNA. (2) DNA.

(3) *m*RNA. (4) *r*RNA.

148. Which one among the following doesn't occur during resting phase of cell cycle?

(1) Centriole duplication.

(2) DNA synthesis.

(3) Organelle duplication.

(4) Splitting of centromere.

149. *ss*RNA is observed in all except,

(1) Viroids. (2) QBbacteriophage.

(3) TMV. (4) φ × 174 bacteriophage.

150. Read the following statements.

A. Air currents help in pollen transfer in gymnosperms and grasses.

B. Agarose gel electrophoresis is employed to check the progression of restriction enzyme digestion.

C. The stage between two meiotic divisions is called interkinesis and generally long with DNA replication.

D. In both chloroplast and mitochondria, ATP is produced from ADP and inorganic phosphate in the presence of light.

Choose the correct pair of statements.

(1) A and B. (2) B and C.

(3) C and D. (4) A and D.

151. What is the ratio of gross ATP produced (SLP) in Glycolysis and Kreb's cycle if one molecule of glucose undergoes oxidation?

(1) 1 : 1 (2) 1 : 2

(3) 2 : 1 (4) 3 : 1

152. Which one among the following is ***not*** observed in different algal members?

(1) Different types of chlorophyll pigments (*a, b, c,* and *d*)

(2) Cell wall with pectocellulose, algin, galactans, mannans and agar.

(3) Gametic fusion may be isogamous, anisogamous (or) oogamous.

(4) Thallus is non-green, heterotrophic and largely terrestrial with restricted form and size.

153. Given below are two statements.

Statement I : In *Pinus,* pollen tube carries male gametes and discharge their contents near the mouth of archegonia.

Statement II : In *Pisum,* pollen tube carries two male gametes and release them into the cytoplasm of the synergid.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

154. Match List–I with List–I.

	List–I		List–II
A.	ATP synthase	**I.**	Oxidative decarboxylation of 3C end product of glycolysis
B.	PEP *case*	**II.**	Provide channel for the movement of protons across the membrane from lumen to stroma and subsequent synthesis of ATP
C.	RubisCO	**III.**	Catalyse binding of 5C keto-sugar to oxygen and help in synthesis of phosphoglycolate and phosphoglycerate
D.	Pyruvic dehydrogenase	**IV.**	Catalyse binding of 3C organic compound with CO_2 in mesophyll cells

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	II	III	IV	I
(2)	II	IV	III	I
(3)	I	II	III	IV
(4)	IV	III	II	I

155. Sex organs (Antheridia and Archegonia) in *Funaria* are present on

(1) Prothallus. (2) Strobilus.

(3) Capsule. (4) Leafy shoot.

156. "All living organisms retain the enzyme machinery to partially oxidise glucose both in aerobic and anaerobic conditions". The above statement is applicable to

(1) Glycolysis. (2) Link reaction.

(3) Krebs cycle. (4) Fermentation.

157. Which one among the following is a complex polysaccharide ?

(1) Starch

(2) Glycogen

(3) Inulin

(4) Chitin

158. Which process among the following occurs inside nucleus?

(1) Translation of *m*RNA into protein.

(2) Centriole duplication in animal cells.

(3) Splicing, capping and tailing.

(4) ATP production by chemiosmosis

159. Which factor, mostly in sub-optimum level, limits photosynthetic rate?

(1) CO_2

(2) Light.

(3) Oxygen.

(4) Temperature.

160. Pick the ***incorrect*** match.

(1) *Porphyra* : Lack flagella throughout the life.

(2) *Alternaria* : Lack sexual reproduction throughout life.

(3) Slime moulds: Lack cell wall throughout its life cycle .

(4) *Selaginella* : Lack seeds throughout its life cycle.

161. Pick the ***incorrect*** match.

(1) Seed germination inhibitor : GA.

(2) Succinic dehydrogenase inhibitor : Malonate.

(3) Cholesterol synthesizing enzyme inhibitor : Statins.

(4) Preventing RNA polymerase from binding to promoter in *lac* operon : Repressor protein.

162. Linnaeus classified flowering plants based on a criterion namely

(1) Nutrition.

(2) Presence/absence of cell wall.

(3) Androecium structure.

(4) Nature of stem.

163. Mendel's law which is universally true in all cases is

A. Law of dominance

B. Law of segregation

C. Law of independent assortment

(1) only A (2) only B

(3) A and B only (4) A and C only

164. Which one among the following is ***not*** a nitrogen fixer, instead, help in increased phosphate absorption from soil?

(1) *Glomus.*

(2) *Rhizobium.*

(3) *Azotobacter.*

(4) *Azospirillum.*

165. Two protein discs on the primary constriction of chromosomes to which spindle fibres attach during cell division are

(1) Centrioles. (2) Telomeres.

(3) Satellite. (4) Kinetochores.

166. Pick the ***incorrect*** match.

(1) *Ori*: origin of DNA replication.

(2) Promoter: start site of transcription.

(3) UGA: terminates translation.

(4) Sigma factor: terminates transcription.

167. Given below are two statements.

Statement I : There could be many transition states in between substrate and product in enzyme catalyzed reaction.

Statement II : Binding of substrate induces the enzyme to alter its shape during enzyme catalysis.

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct, but statement II is incorrect.

(4) Statement I is incorrect, but statement II is correct.

168. In which of the two steps given below, activity of endonuclease and ligase is observed?

(1) *r*DNA synthesis and crossing over.

(2) *r*DNA synthesis and PCR.

(3) PCR and downstream processing.

(4) PCR and gel electrophoresis.

169. Pick the ***incorrect*** match.

(1) Myxotrophic nutrition : *Euglena.*

(2) Saprophytic nutrition : Slime moulds.

(3) Chemosynthetic nutrition : Plants.

(4) Holozoic nutrition : Animals.

170. The former is an exception to degenerate nature of genetic code and the latter is an exception to universality of genetic code. What are the former and the latter ?

(1) Methionine and mitochondrial codons.

(2) Protozoan codons and Tryptophan.

(3) Mitochondrial codons and protozoan codons.

(4) Tryptophan and methionine.

171. During development of callus into plantlet if cytokinins and auxins are added then which among the following is observed?

(1) Morphogenesis.

(2) Apical dominance.

(3) Bolting.

(4) Thinning.

172. The breakdown of hydrogen bonds in a DNA helix during DNA replication, transcription and PCR is initiated by (same order)

(1) Helicase, helicase, and 92°C

(2) Helicase, helicase, and helicase

(3) Helicase, RNA polymerase, and 92°C

(4) RNA polymerase, helicase, and 92°C

173. Which amino acid is produced in mesophyll cells during C_4 pathway?

(1) Glycine.

(2) Serine.

(3) Aspartic acid.

(4) Glutamic acid.

174. Match List–I with List–I.

	List–I		List–II
A.	Monoadelphy & Reniform anthers	**I.**	*Laminaria (Brown alga)*
B.	Frond is edible	**II.**	*Dryopteris (fern)*
C.	Pinnately compound leaves	**III.**	*Marchantia (Liver wort)*
D.	Gemmae	**IV.**	*Hibiscus (Malvaceae)*

Choose the correct answer from the options given below.

	(A)	(B)	(C)	(D)
(1)	IV	I	II	III
(2)	I	IV	II	III
(3)	I	II	IV	III
(4)	IV	III	II	I

175. Thick lignified walled dead conducting elements with tapering ends without any terminal perforation is '**A**' and those which are thick lignified walled cylindrical conducting elements with terminal perforation is '**B**'. **A** and **B** respectively, are

	A	**B**
(1)	Sieve cells	Sieve tubes
(2)	Sieve tubes	Sieve cells
(3)	Tracheids	Vessels
(4)	Vessels	Tracheids

176. Given below are two statements.

Statement I : Individual folded polypeptides in quaternary structure could be linear string of spheres or spheres arranged one upon each other in the form of cube or plate.

Statement II : Haemoglobin is a protein with quaternary structure with four polypeptides.

In light of the above statements, choose the correct answer from the options given below.

(1) Statement II is correct, but statement I is incorrect.

(2) Both the statements I and II are correct.

(3) Statement I is correct, but statement II is incorrect.

(4) Both statements I and II are incorrect.

177. Read the following floral characters of a flower.

A. Petals : Tubular corolla

B. Stamens : Syngenecious and epipetalous

C. Carpels : Two, inferior ovary with a single ovule on basal placentation.

D. Fruit : Cypsela

E. Seed : Non endospermic

The above characters are pertinent to

(1) Sunflower (2) Mustard

(3) China rose (4) Lily

178. Synaptonemal complex dissolution and formation of chaismata are the events occurring in

(1) Diplotene (2) Anaphase I

(3) Telophase I (4) Leptotene

179. Unicellular eukaryote with a chitinous cell wall is

(1) *Bacillus.*

(2) *Euglena.*

(3) *Gonyaulax.*

(4) *Saccharomyces* (yeast).

180. Which metabolic pathway is ATP and temperature dependent, but light independent?

(1) Non-cyclic phosphorylation.

(2) Photorespiration.

(3) Calvin cycle.

(4) Krebs cycle.

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK

SPACE FOR ROUGH WORK